

TrES-3b

R-2578

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110320 · created 2026-08-27T00:34:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455639.69795 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.012
Transit depth (Rp/Rs)^2	3.86 +/- 0.48 [%]
Orbital Inclination (inc)	83.51 +/- 0.82
Airmass coefficient 1 (a1)	1.134 +/- 0.015
Airmass coefficient 2 (a2)	-0.107 +/- 0.062
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#2 - [498, 401]
Optimal Aperture	3.37
Optimal Annulus	5.25
Transit Duration (day)	0.0324 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.012 vs 0.16309 (deviation 2.41σ)
Duration fit vs published	0.0324 d vs 0.0567 d (deviation 7.97σ)
Mid-time O–C	-1879.5 min
Depth SNR	14.0
Residual scatter	1.35 %

Light curve

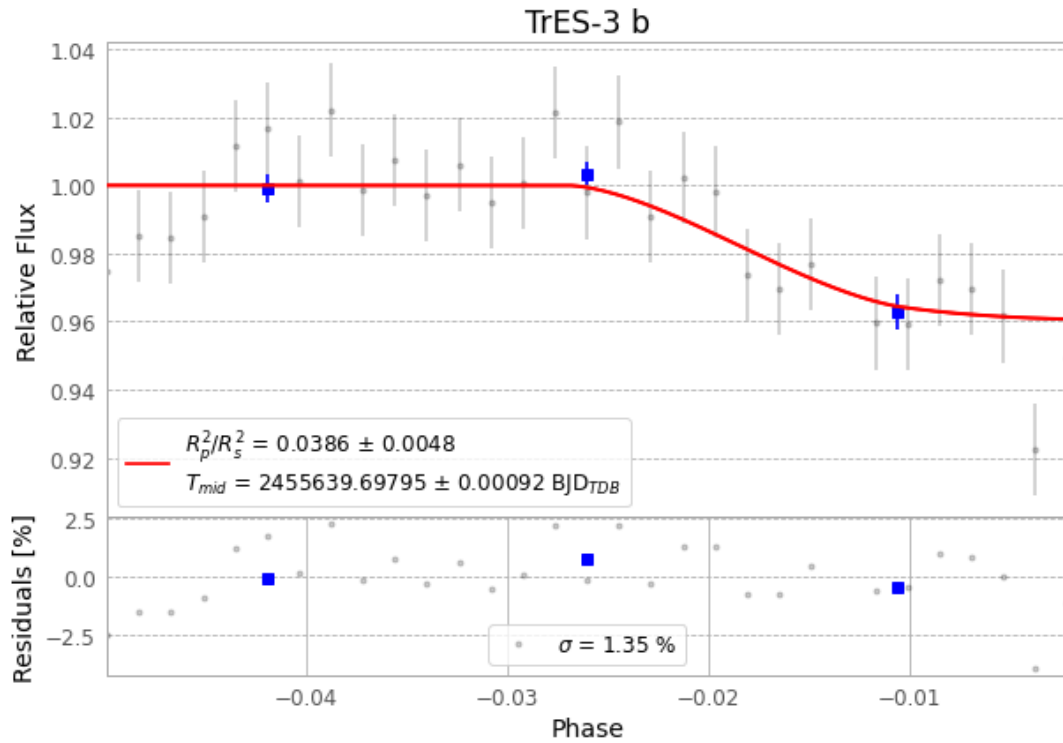


Figure 1 — FinalLightCurve_TrES-3 b_2011-03-20.png (SHA-256 c31f2cb927691b36...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2867f313a3b1c9ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

31 FITS frames (20 MB), TRES-3110320103013.FITS → TRES-3110320120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110320.log (cca2fe83f4c4...), FinalLightCurve_TrES-3 b_2011-03-20.png (c31f2cb92769...), FinalLightCurve_TrES-3 b_2011-03-20.csv (c64b4345f881...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:34Z.